

Introduction to Exploratory Data Analysis

Course 3. Outlier detection

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Outliers

*“An outlier is an observation that **deviates** so much from other observations as to arouse suspicion that it was **generated by a different mechanism**”.*

Douglas Hawkins, *Identification of outliers*, 1980, Chapman and Hall.

Outliers vs Anomalies

Outliers

- From statistics
- Used in the context of a probability distribution

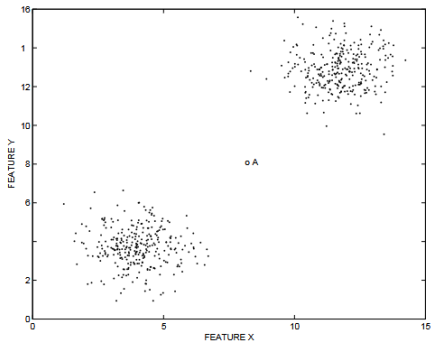
Anomalies

- From medicine, then machine learning & signal processing
- Can imply a value judgment (undesirable, faulty, new)

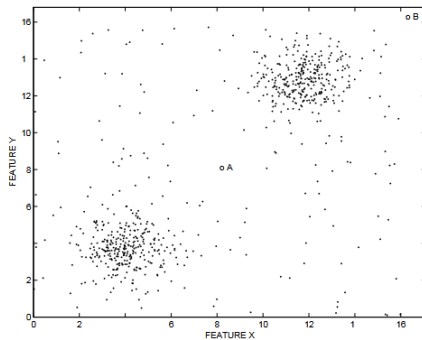
Both

- Few samples are outliers/anomalies
- Limit between normal and abnormal is context-dependent
- Hard to distinguish from noise (and sometimes from normal)

Outliers

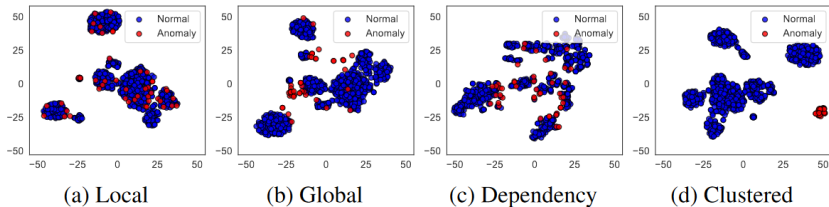


(a)



(b)

Anomalies



Why do outlier analysis / anomaly detection

1. As a data cleaning step

- Leans more on the outlier identification side
- Outliers hinder data analysis (i.e. garbage data, errors in data acquisition, aka artifacts)
- Tends to view outliers as extreme values
- Considered meaningless, to be excluded from further analysis

2. To understand the data

- Leans more on the anomaly detection side
- Anomalies are part of data characterization (e.g. may indicate critical issues)
- Considered meaningful, to be included in further analysis

Why eliminate outliers

They do not describe the process (are errors).

Some methods are not robust to outliers.

Consider the following one-dimensional dataset:

$\mathbf{x} = [2.993, 3.006, 2.999, 2.998, 3.001, 3.003]$.

We wish to model the data and estimate the mean and variance.

$$\mu = 3$$

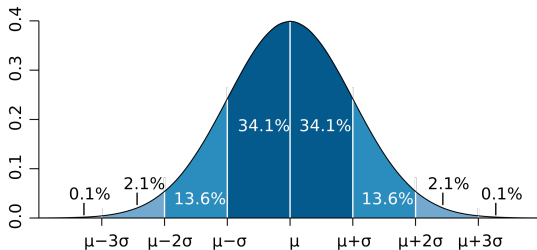
$$\sigma = 0.0044$$

Add one outlier $\mathbf{x} = [2.993, 3.006, 2.999, 2.998, 2845, 3.001, 3.003]$.

$$\mu = 409$$

$$\sigma = 1074$$

Extreme values



Z score (Z-value) for extreme value analysis

$$Z_i = \frac{|X_i| - \mu}{\sigma}, \quad (1)$$

where μ is the mean and σ is the standard deviation.

Computes the number of standard deviations a value is away from the mean.

OBS 1. The score needs to be interpreted: how many standard deviations from the mean constitute an outlier?

The choice may arbitrary.

If no clue, choose $Z\text{-value} = 3$.

OBS 2.

Let $X = [1, 1, 1, 2, 2, 50, 50, 98, 98, 99, 99, 99]$.

Will the Z -value identify the outliers?

Z -score can be used for outlier detection if the data is normally distributed.

In general, the probability distribution of the data needs to be known/determined in order to choose the right measure.

Extreme value tests

Better way: methods that do not depend on data distribution.

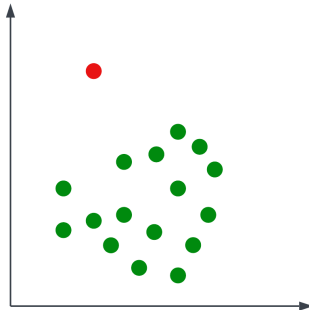
Chebyshev inequality

$$P(|x - \mu| \geq k\sigma) \leq \frac{1}{k^2} \quad (2)$$

Markov inequality

$$P(x \geq a) \leq \frac{\mu}{a} \quad (3)$$

1. Anomalies are responsible for most of the variance in the data.
2. Anomalies lie on the margins of data cloud.
3. Anomalies are found in lower-density regions of the data.
4. Anomalies manifest in some sub-space of the data.
5. Anomalies are easier to separate from the rest of the data than normal samples are.
6. If data is modeled using statistical tools, the model will better represent normal samples and will do worse with outliers.

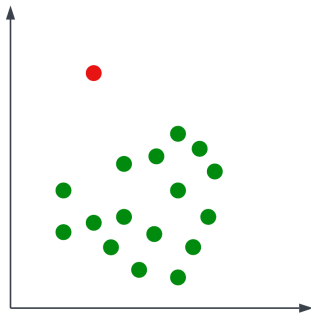


Multidimensional signals

Let $\mathbf{X} \in \mathbb{R}^{n \times d}$ be the dataset with n samples.

Let $\mathbf{x}_i \in \mathbb{R}^d$ be the i -th sample, where $i \in [1, n]$.

Let $\mathbf{x}$ be a sample (a line in $\mathbf{X}$).



HBOS

Histogram-Based Outlier Score

How likely is it for a sample $\mathbf{x}$ to exist?

$$p(\mathbf{x}) = \prod_{i=1}^d p_i(x_i) \quad (4)$$

where x_i is the value of the sample on feature i and

$$p_i(x_i) = \frac{1}{N} \frac{\text{bin_count}}{\text{bin_width}} \quad (5)$$

where N is the number of samples in the dataset.

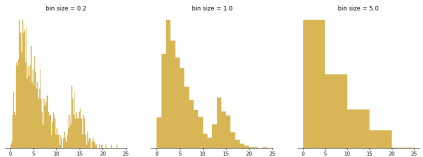
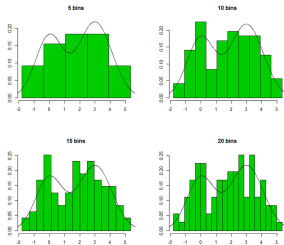
Here, $p_i(x_i)$ is the height of each bin, i.e. the density of the bin.

HBOS

Bins with a small number of samples mostly contain anomalies.

Choice of number of bins, k , matters.

If no info is available about the data distribution, $k = \sqrt{N}$



HBOS

Create the histogram for *each* dimension.

Compute an outlier score associated with each feature.

a) Small probability, high score.

b) Since product of small numbers leads to numerical problems, take the logarithm: $\log p(x) = \sum_{i=1}^d \log p_i(x_i)$.

$$HBOS(\mathbf{x}) = \sum_{i=0}^d \log\left(\frac{1}{hist_i(\mathbf{x})}\right), \quad (6)$$

where $hist_i(\mathbf{x})$ is the height of the bin in which x is found in each data dimension.

Evaluating performance

Anomaly detection can be viewed as a binary classification problem
 $\implies$ evaluation metrics from classification.

True Positives (TP), False Positives (FP)

True Negatives (TN), False Negatives (FN)

Precision

$$\frac{TP}{TP+FP}$$

Sensitivity, Rate TP (*recall*)

$$\frac{TP}{TP+FN}$$

Specificity, Rate TN

$$\frac{TN}{TN+FP}$$

FP Rate

$$\frac{FP}{FP+TN} = 1 - \textit{specificity}$$

Accuracy

$$\frac{TP+TN}{TP+TN+FP+FN}$$

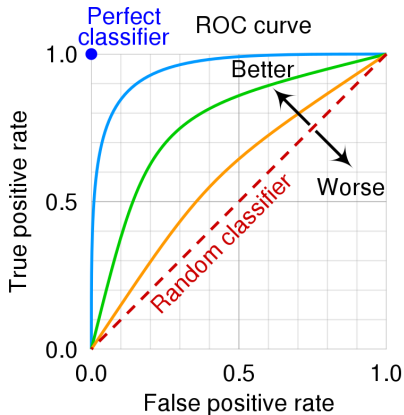
F1 Score

$$2 \frac{\textit{precizie} * \textit{senzitivitate}}{\textit{precizie} + \textit{senzitivitate}} = \frac{2TP}{2TP+FP+FN}$$

Evaluating performance

Receiver Operating Characteristic (ROC)

Graphic of TP rate w.r.t. FP rate.



Evaluating performance

Area under ROC (ROC AUC)

ROC AUC = 1 perfect detection.

What does ROC AUC = 0.5 mean?

Multidimensional signals

Low dimensional spaces: All dimensions are relevant and distances (such as Euclidian distance) can give clues on a sample's normality.

High dimensional spaces: Data points are equidistant. For each data point, the distance between it and the nearest data point is similar to the distance to the furthest point.

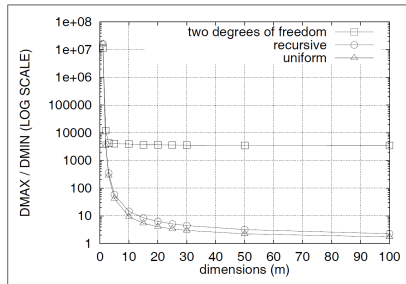


Fig. 7. Correlated distributions, one million tuples.